

SUPPLEMENTARY INFORMATION

Predicting the absence of WHO fungal priority pathogens in agricultural soils using machine learning and ITS metabarcoding

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Supplementary Materials and Methods

Feature Set

The input features included drought level, soil water content, soil temperature, organic matter, nitrogen, phosphorus, potassium, and soil pH as continuous variables. Crop type was the only categorical variable and was encoded to one-hot variables for compatibility with the model. To evaluate the influence of different combinations of environmental parameters, four feature sets (FS1-FS4) were defined (Table S2). Feature Set 1 (FS1) comprised all measured parameters, including crop type. Feature Set 2 (FS2) included all parameters except crop type, which was used to assess whether absence conditions were consistent across crop types. Feature Set 3 (FS3) excluded both crop type and soil temperature, and Feature Set 4 (FS4) excluded only soil temperature. The exclusion of soil temperature in FS3 and FS4 was due to a lack of soil temperature measurement in the independent test samples, ensuring that the resulting absence conditions could be applied directly to those samples without imputation. These feature sets were chosen on a combination of biological relevance, practical constraints, and the need to evaluate robustness across different variable subsets. The use of the identified feature sets enabled testing of whether absence conditions remained consistent with potentially influential parameters, such as crop type or soil temperature, were removed, while still aiming to achieve high predictive precision and recall.

Hyperparameter Tuning

To ensure reproducibility and to control model complexity, hyperparameter selection for the decision tree models was performed using stratified k-fold cross validation where feasible. Stratification was used to preserve class proportions across folds, which is recommended for imbalanced data (Hastie et al. 2001). Since the number of absence samples varied among pathogens (Table 1), a single cross-validation configuration was not appropriate for all cases. Therefore, the cross-validation for hyperparameter tuning was conducted separately for each pathogen and for each feature set, resulting in distinct parameter configurations for each pathogen–feature set combination.

Decision tree complexity was controlled through two key hyperparameters: (1) the maximum tree depth (*max_depth*), which limits the number of recursive splits and constrains model complexity, and (2) the minimum number of samples required in a terminal node (*min_samples_leaf*), which controls the minimum leaf size and affects model sensitivity to small sample partitions. These parameters are widely

used to regularize decision tree models and improve generalization by limiting overly specific decision boundaries (Breiman et al. 1984; Hastie et al. 2001; Kuhn and Johnson 2013). To limit the degrees of freedom during model selection and reduce the risk of overfitting from extensive search in small and imbalanced datasets, hyperparameter tuning was restricted to a small, predefined grid with *max_depth* values of 2, 3, and 4, and *min_samples_leaf* values of 1, 2, and 3 (Hastie et al. 2001; Pedregosa et al. 2011). The *random_state* value was fixed at 42 for all model runs to ensure reproducibility.

Hyperparameter tuning was conducted separately for each feature set, which allows model complexity to adapt to different combinations of environmental variables. For pathogens with sufficient minority-class samples, hyperparameters were selected using stratified k-fold cross-validation by maximizing the absence-class F1-score, which balances precision and recall (Hastie et al. 2001). The hyperparameter combination yielding the highest mean absence-class F1-score across cross-validation folds was selected for each pathogen-feature set pair. The number of folds was chosen to ensure that each validation fold contained at least two minority-class samples to avoid unstable splits (Hastie et al. 2001; Kuhn and Johnson 2013). Depending on pathogen-specific sample availability, the number of folds (*k*) ranged from 2 to 5. Specifically, *k* = 5 was used for *Rhizopus* spp., *Scedosporium* spp., and *Talaromyces marneffe*; *k* = 3 was used for *Acremonium* spp., *Candida tropicalis*, and *Falciformispora senegalensis*; and *k* = 2 was used for *Mucor* spp. For pathogens with a limited number of minority classes (≤ 3), including *Curvularia lunata* and *Lichtheimia* spp., cross-validation was not performed as fold-based estimation would be statistically unreliable (Domingos 2012). For these two pathogens, conservative fixed hyperparameter values (*max_depth* = 3; *min_samples_leaf* = 2) were used to reduce overfitting and to ensure interpretability.

Cross-validation was used exclusively for hyperparameter selection, while final model performance was assessed using an independent test dataset to avoid optimistic bias, see Validation of Identified Absence Conditions section.

Supplementary Results

Characterization of Environmental Parameter Distributions in the Training Dataset

Systematic differences between measured environmental parameters of pathogen-present and pathogen-absent samples were evaluated by quantifying effect sizes using standardized mean

differences (SMD). This analysis provides an empirical, model-independent assessment of the magnitude and direction of separation between the two groups, which allows identification of environmental variables that differ most strongly between pathogen presence and absence. SMD values were calculated for each environmental parameter in the training dataset. The environmental parameters analyzed included both continuous variables and categorical crop types, with the latter encoded as binary indicators. A summary of SMD values across all pathogen-feature pairs is presented in Figure 1. Corresponding mean and standard deviation for each environmental parameter in pathogen-present and pathogen-absent training samples are provided in Table S4.

Several environmental features exhibited large separation between pathogen-present and pathogen-absent samples, with absolute SMD values exceeding 0.8 in multiple cases (Andrade 2020). This indicates that certain environmental conditions differ notably between two classes. Negative SMD values indicate that the average feature value was higher in pathogen-absent samples, and positive values indicate higher values in pathogen-present samples. Thus, both the sign and magnitude of the SMD provide interpretable insights into environmental conditions associated with absence of each pathogen.

For example, phosphorus levels were higher in samples where *Lichtheimia* spp. was not detected (SMD = -1.59), suggesting a potential association between elevated phosphorus and reduced pathogen occurrence (Figure 1). In contrast, samples with *Mucor* spp. showed strong association with higher organic matter (SMD=1.08), soil water content (SMD = 0.71), and potassium (SMD = 0.63). The lower values of these three parameters may be indicative of environmental conditions where *Mucor* spp. is absent. *Curvularia lunata* was more frequently present in samples with higher soil temperature (SMD = 1.62), and *Falciformispora senegalensis* was more often detected under wetter conditions, as reflected by a negative SMD for drought level (SMD = -0.72) and positive SMD for soil water content (SMD = 0.69).

Overall, environmental parameters such as organic matter and soil water content showed large effect sizes across multiple pathogens, which indicates their potential importance in distinguishing pathogen presence and absence. Features exhibiting large SMD values do not always correspond to those selected by the decision tree models, which highlights that multivariate absence conditions cannot be inferred from univariate effect sizes alone. This distinction emphasizes the necessity of the decision

tree approach for identifying actionable combinations of environmental thresholds rather than isolated variables. As a model-independent, univariate method, SMD provides a measure of which parameters differ most between the two classes and facilitates interpretation of the multivariate absence conditions identified by the decision tree models. The role of these parameters in defining the specific environmental conditions associated with pathogen absence is further examined in the following section using decision tree models.

References

- Andrade C (2020) Mean Difference, Standardized Mean Difference (SMD), and Their Use in Meta-Analysis: As Simple as It Gets. *Journal of Clinical Psychiatry* 81:. <https://doi.org/10.4088/JCP.20F13681>
- Breiman L, Friedman JH, Olshen RA, Stone CJ (1984) *Classification and regression trees*. CRC Press, New York
- Domingos P (2012) A few useful things to know about machine learning. *Commun ACM* 55:78–87. <https://doi.org/10.1145/2347736.2347755>
- Hastie T, Tibshirani R, Friedman J (2001) *The Elements of Statistical Learning*. Springer New York Inc., New York, NY, USA
- Kuhn M, Johnson K (2013) *Applied Predictive Modeling*, 1st edn. Springer, New York
- Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12:2825–2830

Supplementary Tables

Table S1. Distribution of number of samples for pathogen presence and absence in the training dataset and the test dataset.

Pathogen	WHO Priority Level	Training Dataset		Test Dataset	
		Absence	Presence	Absence	Presence
<i>Acremonium</i> spp.	High	9	112	11	1
<i>Candida tropicalis</i>	High	112	9	12	0
<i>Curvularia lunata</i>	High	118	3	12	0
<i>Falciformispora senegalensis</i>	High	112	9	12	0
<i>Fusarium</i> spp.	High	0	121	0	12
<i>Lichtheimia</i> spp.	High	118	3	12	0
<i>Mucor</i> spp.	High	117	4	12	0
<i>Rhizopus</i> spp.	High	89	32	8	4
<i>Scedosporium</i> spp.	Medium	59	62	1	11

<i>Talaromyces marneffe</i>	Medium	87	34	12	0
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Table S2. Summary of the four feature sets.

Feature Set	Parameters included	Parameter excluded
FS1	drought level, soil water content, organic matter, nitrogen, phosphorus, potassium, soil temperature, soil pH, crop type	None
FS2	drought level, water soil content, organic matter, nitrogen, phosphorus, potassium, soil temperature, soil pH	crop type
FS3	drought level, soil water content, organic matter, nitrogen, phosphorus, potassium, soil pH	soil temperature, crop type
FS4	drought level, soil water content, organic matter, nitrogen, phosphorus, potassium, soil pH, crop type	soil temperature

Table S3. The environmental data comparison between 2022 and 2025.

March 2022 (*n* = 11)

Sample code	Zone	District	Drought level	Crop	Soil Water_Content (%)	Organic_matter	N_Nitrate	N_Ammonium	Phosphorus	Potassium	Soil pH	Soil temperature (°c)
RK1	four	Phu phan	1	Rubber tree A	11.73	1.5	0		1	1	6.8	29.5
RK2	four	Phu phan	1	Rubber tree A	10.90	1.5	0		1	1	6.9	30.7
RB1	five	Ban mueang	1	Rubber tree B	9.95	3.5	1		1	1	7	28.5
RB2	five	Ban mueang	1	Rubber tree B	9.73	3.5	0		1	1	7	29.2
RB3	five	Ban mueang	1	Rubber tree B	10.75	3.5	0		1	0	7	28.2

CK1	four	Phu phan	1	Cassava	17.32	0.5	0		1	3	6	28.2
CK2	four	Phu phan	1	Cassava	15.27	0.5	0		1	2	6.9	28.5
CK3	four	Phu phan	1	Cassava	15.94	0.5	0		1	2	6.8	28.2
SS1	five	Song Dao	1	Sugarca ne	8.36	2.5	1		1	0	6.6	31.7
SS2	five	Song Dao	1	Sugarca ne	8.09	1.5	1		1	0	6.5	33.4
SS3	five	Song Dao	1	Sugarca ne	7.99	2.5	1		1	1	6.1	30.8

February 2025 (*n* = 12)

Sample code	Zone	District	Drought level	Crop	Soil Water_ Content (%)	Organic _matter	N_ Nitrate	N_ Ammonium	Phosph orus	Potass ium	Soi l pH	Soil temperat ure (°c)
RK1	four	Phu phan	2	Rubber tree A	2.56	2.5	0	2	2	1	6.5	
RK2	four	Phu phan	2	Rubber tree A	2.12	2.5	0	1	1	1	6.5	
RK3	four	Phu phan	2	Rubber tree A	3.61	2.5	0	1	1	1	6.5	
RB1	five	Ban muean g	1	Rubber tree B	9.61	2.5	0	1	0	1	6.5	
RB2	five	Ban muean g	1	Rubber tree B	3.66	2.5	0	1	0	1	5.5	
RB3	five	Ban muean g	1	Rubber tree B	8.56	2.5	0	2	0	1	6.5	

CK1	four	Phu phan	2	Cassava	1.80	1.5	1	1	2	1	6
CK2	four	Phu phan	2	Cassava	4.41	1.5	0	1	0	1	6
CK3	four	Phu phan	2	Cassava	1.89	1.5	0	1	0	1	6
SS1	five	Song Dao	3	Sugarca ne	4.44	1.5	0	1	1	1	5.5
SS2	five	Song Dao	3	Sugarca ne	2.63	1.5	0	0	2	1	6
SS3	five	Song Dao	3	Sugarca ne	5.35	1.5	0	2	2	1	5.5

Table S4. Summary statistics including mean $\pm$ standard deviation of environmental parameters for pathogen-negative (class 0) and pathogen-positive (class 1) samples in the training dataset.

Pathogen	Class	Drought	Soil Water Content	Organic Matter	N	P	K	Soil Temp (°c)	Soil pH
<i>Acremonium</i> spp.	0	1.89 $\pm$ 1.17	3.49 $\pm$ 5.34	2.61 $\pm$ 0.78	0.44 $\pm$ 0.53	1.44 $\pm$ 0.73	0.67 $\pm$ 0.50	29.69 $\pm$ 3.34	6.22 $\pm$ 0.73
	1	2.25 $\pm$ 1.19	11.30 $\pm$ 9.75	2.23 $\pm$ 0.97	0.79 $\pm$ 0.81	1.43 $\pm$ 0.97	0.58 $\pm$ 0.69	29.58 $\pm$ 3.19	6.01 $\pm$ 0.87
<i>Candida tropicalis</i>	0	2.17 $\pm$ 1.17	10.78 $\pm$ 9.78	2.26 $\pm$ 0.95	0.77 $\pm$ 0.81	1.42 $\pm$ 0.96	0.55 $\pm$ 0.67	29.57 $\pm$ 3.26	6.07 $\pm$ 0.84
	1	2.89 $\pm$ 1.27	10.00 $\pm$ 9.13	2.28 $\pm$ 1.09	0.78 $\pm$ 0.67	1.56 $\pm$ 0.73	1.00 $\pm$ 0.71	29.73 $\pm$ 2.29	5.54 $\pm$ 1.08
<i>Curvularia lunata</i>	0	2.23 $\pm$ 1.18	10.73 $\pm$ 9.78	2.28 $\pm$ 0.96	0.76 $\pm$ 0.80	1.42 $\pm$ 0.95	0.58 $\pm$ 0.68	29.46 $\pm$ 3.08	6.03 $\pm$ 0.86
	1	2.00 $\pm$ 1.73	10.58 $\pm$ 6.91	1.50 $\pm$ 0.00	1.00 $\pm$ 0.00	1.67 $\pm$ 1.15	0.67 $\pm$ 0.58	34.50 $\pm$ 4.03	5.87 $\pm$ 1.10
<i>Falciformispora senegalensis</i>	0	2.29 $\pm$ 1.20	10.23 $\pm$ 9.72	2.22 $\pm$ 0.96	0.77 $\pm$ 0.81	1.46 $\pm$ 0.98	0.60 $\pm$ 0.69	29.53 $\pm$ 3.01	6.00 $\pm$ 0.87

	1	1.44 ± 0.73	16.88 ± 7.39	2.72 ± 0.83	0.78 ± 0.67	1.11 ± 0.33	0.44 ± 0.53	30.27 ± 5.09	6.38 ± 0.76
	0	2.25 ± 1.19	10.41 ± 9.37	2.23 ± 0.95	0.78 ± 0.80	1.47 ± 0.93	0.60 ± 0.68	29.58 ± 3.15	6.02 ± 0.87
<i>Lichtheimia</i> spp.	1	1.00 ± 0.00	23.10 ± 16.52	3.50 ± 0.00	0.33 ± 0.58	0.00 ± 0.00	0.00 ± 0.00	29.63 ± 5.44	6.37 ± 0.55
	0	2.24 ± 1.19	10.49 ± 9.37	2.23 ± 0.95	0.79 ± 0.80	1.45 ± 0.94	0.57 ± 0.66	29.67 ± 3.20	6.03 ± 0.88
<i>Mucor</i> spp.	1	1.75 ± 0.96	17.40 ± 17.59	3.25 ± 0.50	0.25 ± 0.50	0.75 ± 0.96	1.00 ± 1.15	27.23 ± 1.90	5.88 ± 0.46
	0	2.25 ± 1.16	10.47 ± 10.00	2.31 ± 1.01	0.66 ± 0.74	1.29 ± 0.84	0.57 ± 0.69	29.18 ± 3.05	6.00 ± 0.86
<i>Rhizopus</i> spp.	1	2.16 ± 1.27	11.42 ± 8.94	2.12 ± 0.79	1.06 ± 0.88	1.81 ± 1.12	0.62 ± 0.66	30.71 ± 3.34	6.12 ± 0.89
	0	2.12 ± 1.10	9.33 ± 9.58	2.36 ± 0.90	0.83 ± 0.79	1.27 ± 0.76	0.59 ± 0.62	29.87 ± 2.90	5.94 ± 0.85
<i>Scedosporium</i> spp.	1	2.32 ± 1.26	12.05 ± 9.71	2.16 ± 1.01	0.71 ± 0.80	1.58 ± 1.08	0.58 ± 0.74	29.32 ± 3.44	6.11 ± 0.88
	0	2.13 ± 1.19	10.94 ± 10.21	2.34 ± 0.91	0.64 ± 0.75	1.29 ± 0.90	0.63 ± 0.70	29.66 ± 3.37	6.20 ± 0.70
<i>Talaromyces marneffei</i>	1	2.47 ± 1.16	10.17 ± 8.37	2.06 ± 1.05	1.09 ± 0.83	1.79 ± 0.98	0.47 ± 0.61	29.40 ± 2.71	5.59 ± 1.07

Table S5. Distribution of presence and absence samples by crop type with crop-dependent absence conditions in training and test datasets.

Crop	Dataset	<i>Candida tropicalis</i>		<i>Curvularia lunata</i>		<i>Rhizopus</i> spp.	
		Absence	Presence	Absence	Presence	Absence	Presence
Cassava	training	18	3	20	1	14	7
	test	3	0	3	0	1	2
Rice	training	39	5	44	0	29	15
	test	0	0	0	0	0	0
Rubber tree	training	29	0	29	0	28	1
	test	6	0	6	0	5	1

Sugar cane	training	26	1	25	2	18	9
	test	3	0	3	0	2	1

Table S6. Comparison of WHO FPPL detected in Sakon Nakhon between the 2022 training (n = 11) and 2025 test (n = 12) datasets, showing taxa classified as shared between years or unique to each period. ✓ represents presence; · represents absence.

Species	2022						2025					
	Rubber tree A	Rubber tree B	Cassava	Sugar cane	Category	Existence	Rubber tree A	Rubber tree B	Cassava	Sugar cane	Category	Existence
<i>Aspergillus fumigatus</i>	0.00	0.00	0.00	0.00		×	3.18	1.32	57.25	19.95	shared	✓
<i>Fusarium</i> sp.	0.82	0.00	53.63	23.20		✓	26.14	8.52	30.21	49.86	shared	✓
<i>Fusarium oxysporum</i>	70.61	87.50	34.42	25.04	shared	✓	29.70	41.00	6.27	11.34	shared	✓
<i>Fusarium chlamydosporum</i>	0.00	0.00	0.00	18.80	rare	✓	0.00	0.00	0.00	0.00		×
<i>Fusarium solani</i>	0.00	0.00	0.00	1.10	rare	✓	11.13	2.25	1.79	8.86	shared	✓
<i>Fusarium beomiforme</i>	0.00	0.00	2.04	0.00	rare	✓	0.00	0.00	0.00	0.00		×
<i>Fusarium longipes</i>	0.00	0.13	0.00	0.00	rare	✓	0.00	0.00	0.00	0.00		×
<i>Fusarium neocosmosporiellum</i>	0.00	0.03	0.00	0.11		✓	0.00	0.00	0.00	0.00		×
<i>Fusarium avenaceum</i>	0.00	0.00	0.00	0.00		×	13.24	38.10	0.00	0.01		✓
<i>Fusarium concentricum</i>	0.00	0.00	0.00	0.00		×	14.08	0.43	0.00	0.00		✓

<i>Fusarium equiseti</i>	0.00	0.00	0.00	0.00		×	1.57	2.84	3.83	9.05	shared	✓
<i>Fusarium sambucinum</i>	0.00	0.00	0.00	0.00		×	0.05	1.25	0.00	0.44		✓
<i>Acremonium</i> sp.	4.76	6.68	2.04	1.06	shared	✓	0.00	0.16	0.00	0.00	rare	✓
<i>Acremonium pinkertoniae</i>	19.70	4.67	0.10	1.77	shared	✓	0.00	0.00	0.00	0.00		×
<i>Acremonium furcatum</i>	2.30	0.04	0.00	6.07		✓	0.00	0.00	0.00	0.00		×
<i>Acremonium verruculosum</i>	0.16	0.02	0.10	0.00		✓	0.00	0.00	0.00	0.00		×
<i>Acremonium hennebertii</i>	0.16	0.05	0.00	0.00		✓	0.00	0.00	0.00	0.00		×
<i>Acremonium polychromum</i>	0.00	0.01	0.00	0.11		✓	0.00	0.00	0.00	0.00		×
<i>Acremonium brachypenium</i>	0.00	0.02	0.00	0.00	rare	✓	0.00	0.00	0.00	0.00		×
<i>Acremonium hyalinulum</i>	0.00	0.00	0.10	0.00	rare	✓	0.00	0.00	0.00	0.00		×
<i>Scedosporium boydii</i>	1.48	0.67	7.56	0.00		✓	0.00	0.00	0.00	0.00		×
<i>Scedosporium prolificans</i>	0.00	0.00	0.00	0.11	rare	✓	0.00	0.00	0.00	0.00		×
<i>Scedosporium aurantiacum</i>	0.00	0.00	0.00	0.00		×	0.93	4.12	0.57	0.47	shared	✓
<i>Talaromyces marneffeii</i>	0.00	0.11	0.00	0.00	rare	✓	0.00	0.00	0.00	0.00		×

<i>Falciformispora senegalensis</i>	0.00	0.07	0.00	0.00	rare	✓	0.00	0.00	0.00	0.00	×
<i>Candida tropicalis</i>	0.00	0.00	0.00	21.85	rare	✓	0.00	0.00	0.00	0.00	×
<i>Rhizopus microsporus</i>	0.00	0.00	0.00	0.78	rare	✓	0.00	0.01	0.08	0.01	✓
total	100%	100%	100 %	100%		21	100%	100%	100 %	100%	11

Supplementary Figures

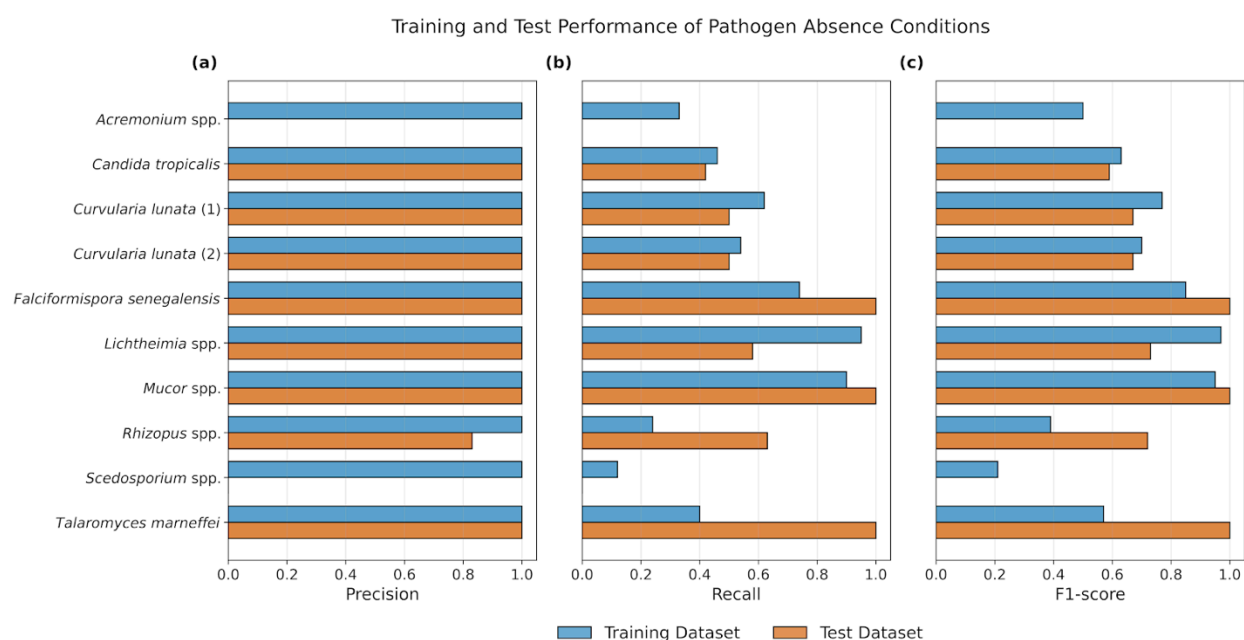


Figure S1. Training and test performance metrics of pathogen absence conditions for nine WHO FPPL pathogens. Panels show precision (a), recall (b), and F1-score (c). High precision indicates reliable identification of pathogen absence. Recall reflects the proportion of true absence samples correctly identified. F1-score summarizes the balance between precision and recall. The corresponding environmental conditions and detailed metrics are provided in Table 1.

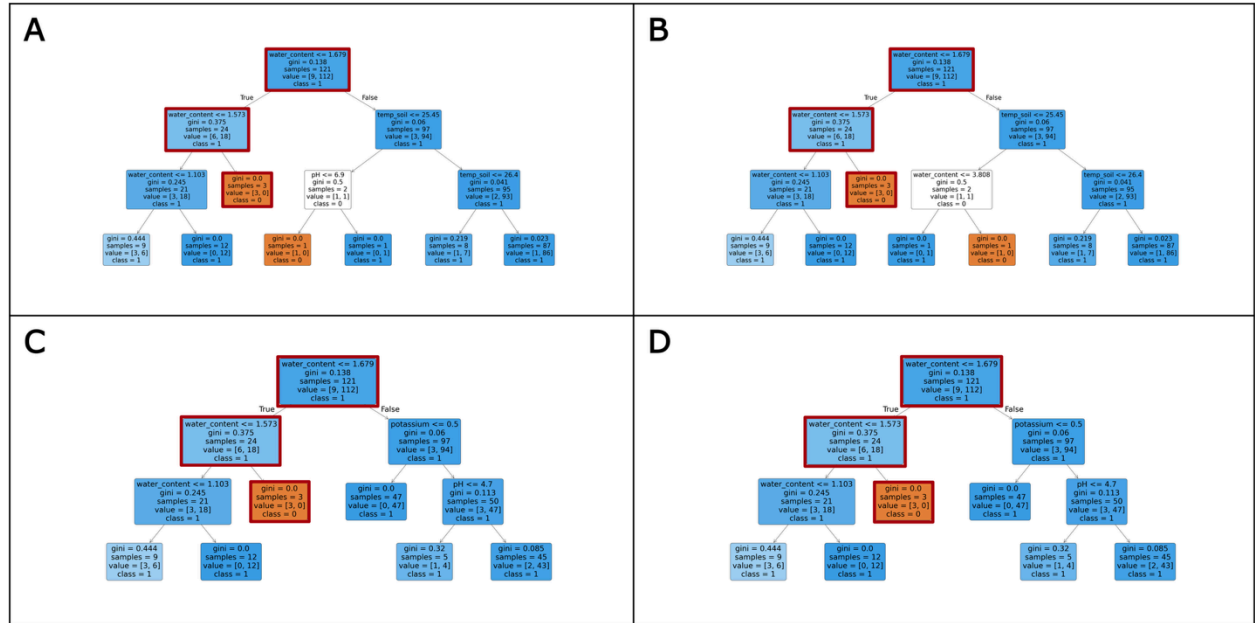


Figure S2. Decision tree models for predicting the presence (class1) or absence (class 0) of *Acremonium* spp. **(A)** Model trained using Feature Set 1 with maximum depth parameter (max_depth) of 3 and minimum samples of terminal leaf node ($min_samples_leaf$) of 2. **(B)** Model trained using Feature Set 2 with maximum depth parameter (max_depth) of 4 and minimum samples of terminal leaf node ($min_samples_leaf$) of 2. **(C)** Model trained using Feature Set 3 with maximum depth parameter (max_depth) of 4 and minimum samples of terminal leaf node ($min_samples_leaf$) of 1. **(D)** Model trained using Feature Set 4 with maximum depth parameter (max_depth) of 3 and minimum samples of terminal leaf node ($min_samples_leaf$) of 2. The root-to-leaf path representing the absence condition of *Acremonium* spp., where soil water content $\leq 1.67945\%$ and soil water content $> 1.57350\%$, is outlined in red. In all sub figures, “gini” indicates the node’s Gini impurity; “samples” is the number of training samples at the node; “value” is the distribution of samples across absence and presence classes; and “class” is the predicted class at that node.

(*min_samples_leaf*) of 2. The root-to-leaf path representing the absence condition of *Curvularia lunata* is outlined in red. Definition of gini, samples, values, class are the same as in Figure S2.

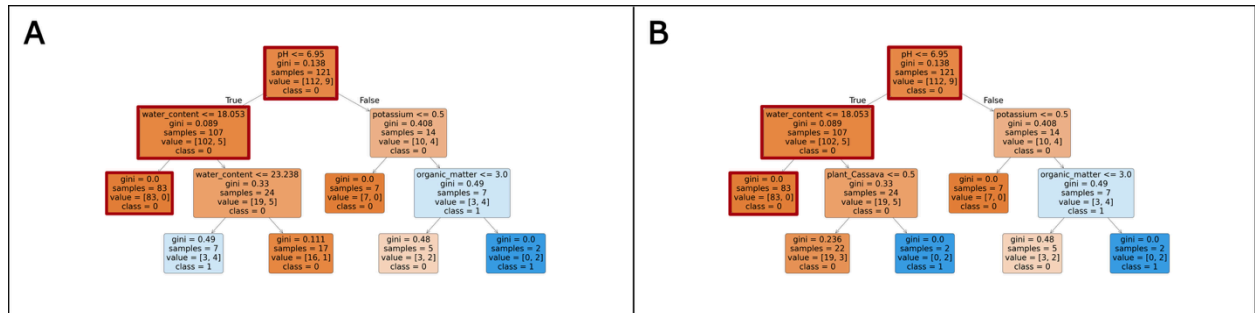
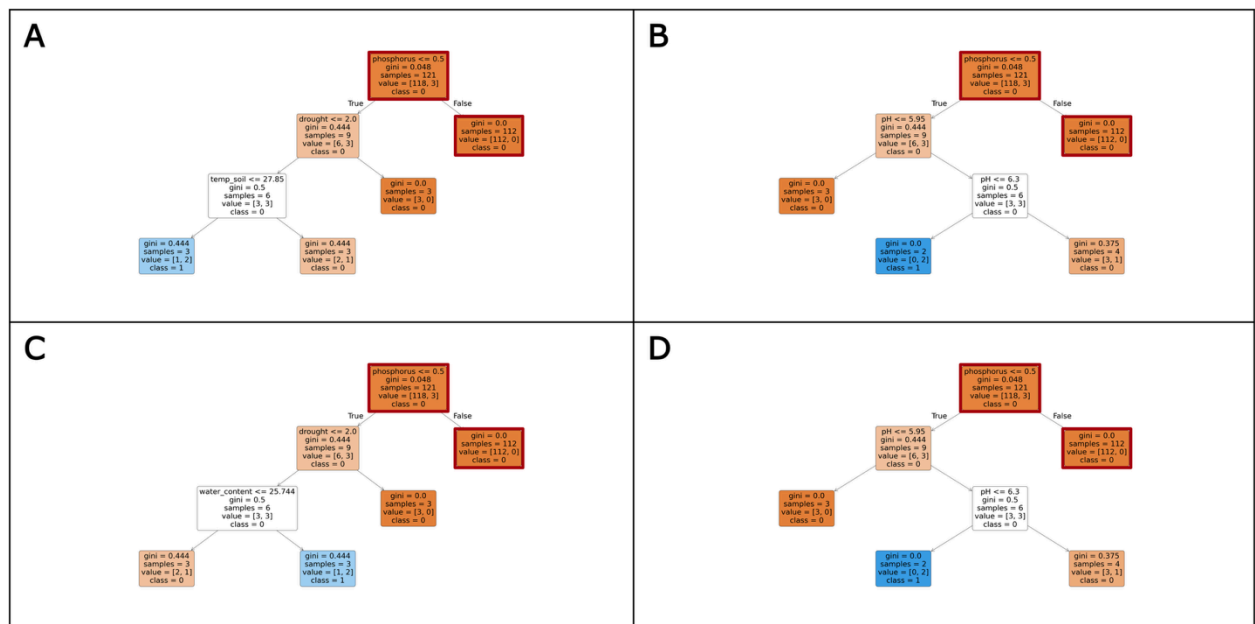


Figure S5. Decision tree models for predicting the presence (class1) or absence (class 0) of *Falciformispora senegalensis*. **(A)** Model trained using Feature Set 3 with maximum depth parameter (*max_depth*) of 4 and minimum samples of terminal leaf node (*min_samples_leaf*) of 1. **(B)** Model trained using Feature Set 4 with maximum depth parameter (*max_depth*) of 4 and minimum samples of terminal leaf node (*min_samples_leaf*) of 2. The root-to-leaf path representing the absence condition of *Falciformispora senegalensis* is outlined in red. Definition of gini, samples, values, class are the same as in Figure S2.



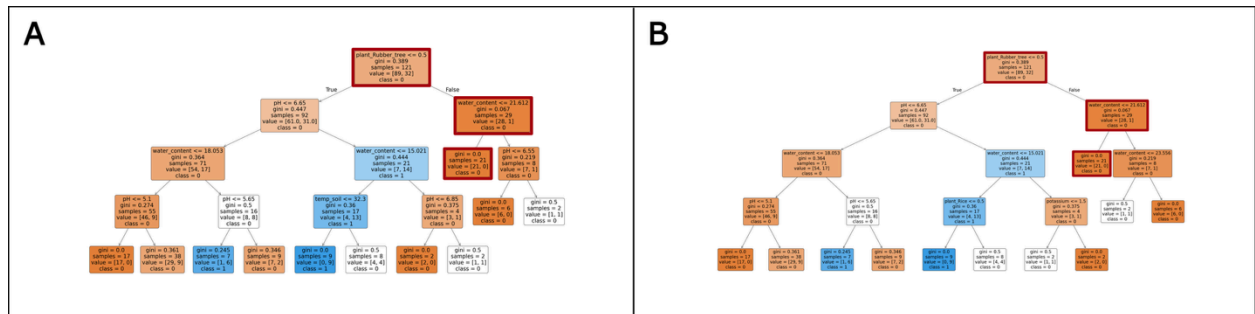


Figure S8. Decision tree models for predicting the presence (class1) or absence (class 0) of *Rhizopus* spp. **(A)** Model trained using Feature Set 1 with maximum depth parameter (*max_depth*) of 4 and minimum samples of terminal leaf node (*min_samples_leaf*) of 2. **(B)** Model trained using Feature Set 4 with maximum depth parameter (*max_depth*) of 3 and minimum samples of terminal leaf node (*min_samples_leaf*) of 2. The root-to-leaf path representing the absence condition of *Rhizopus* spp. is outlined in red. Definition of gini, samples, values, class are the same as in Figure S2.

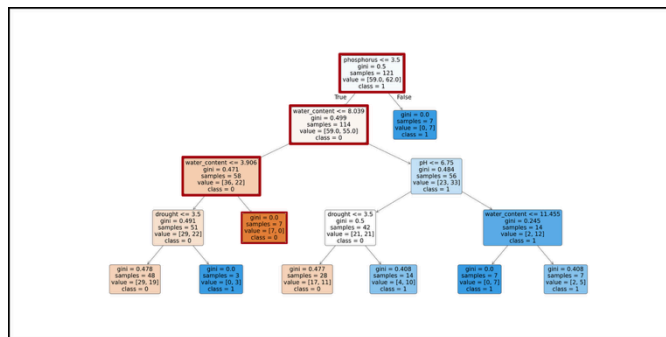


Figure S9. Decision tree models for predicting the presence (class1) or absence (class 0) of *Scedosporium* spp. using environmental parameters from Feature Set 3. The model was constructed using a maximum depth (*max_depth*) of 4 and minimum samples of terminal leaf node (*min_samples_leaf*) of 3. The root-to-leaf path representing the absence condition of *Scedosporium* spp. is outlined in red. Definition of gini, samples, values, class are the same as in Figure S2.

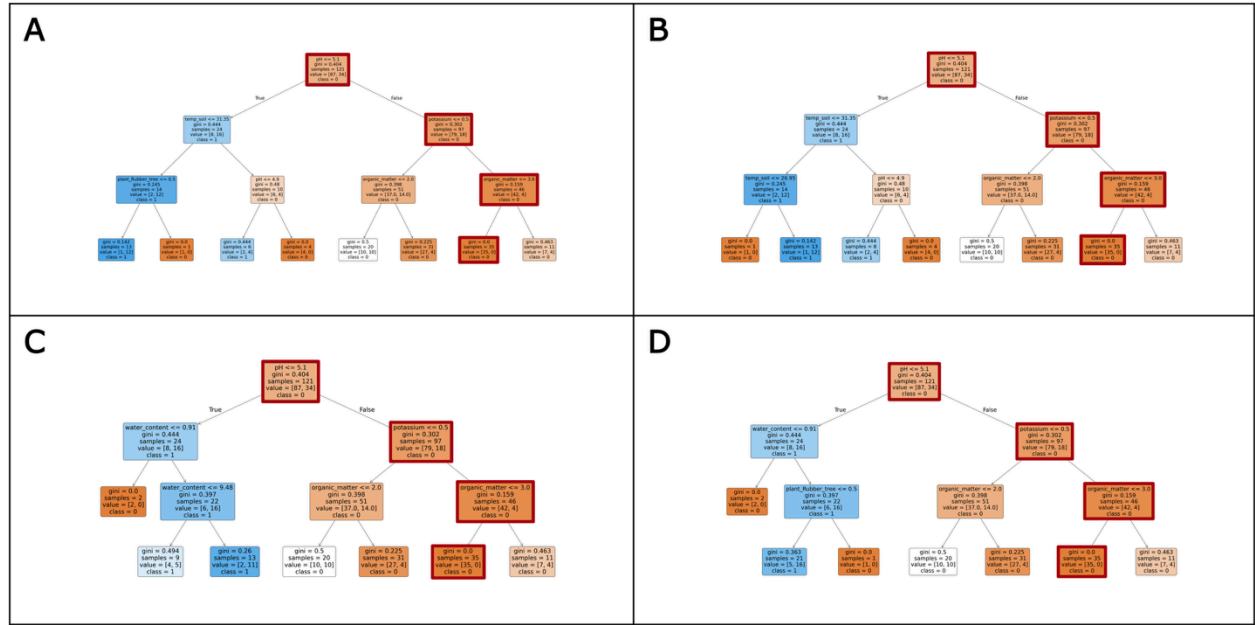


Figure S10. Decision tree models for predicting the presence (class1) or absence (class 0) of *Talaromyces marneffeii* using environmental parameters from Feature Sets 1, 2, 3, and 4, shown in subplots (A), (B), (C), and (D) respectively. All models were constructed using a maximum depth (*max_depth*) of 4 and minimum samples of terminal leaf node (*min_samples_leaf*) of 3. The root-to-leaf path representing the absence condition of *Talaromyces marneffeii* is outlined in red. Definition of gini, samples, values, class are the same as in Figure S2.

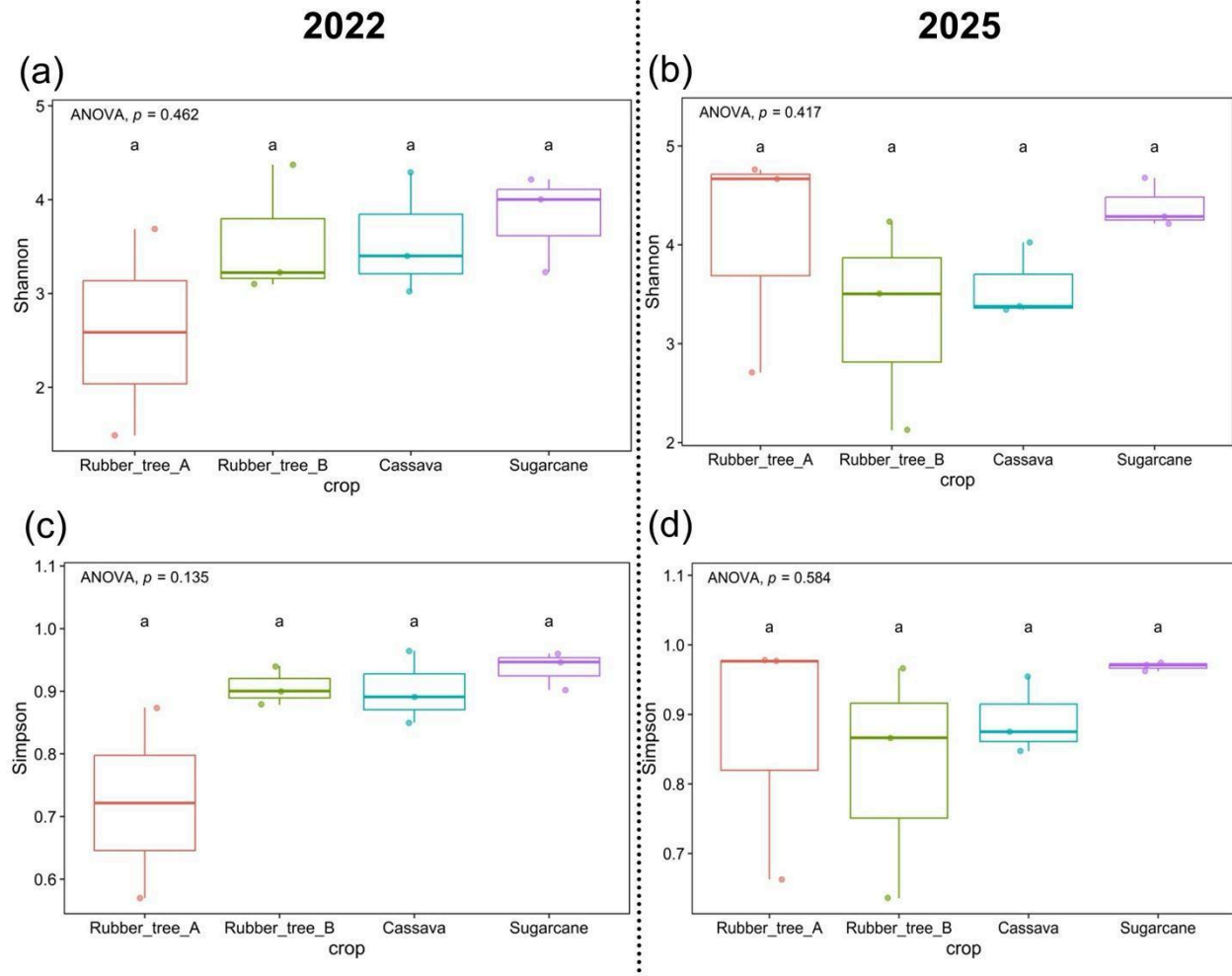


Figure S11. Shannon and Simpson alpha diversity across crop systems from samples collected in 2022 and 2025. (a, b) Shannon and (c, d) Simpson diversity for 2022 (a, c) and 2025 (b, d) across four crop systems. Boxplots show interquartile ranges with median lines; points represent individual samples. No significant differences were observed among crops (ANOVA, $p > 0.05$).

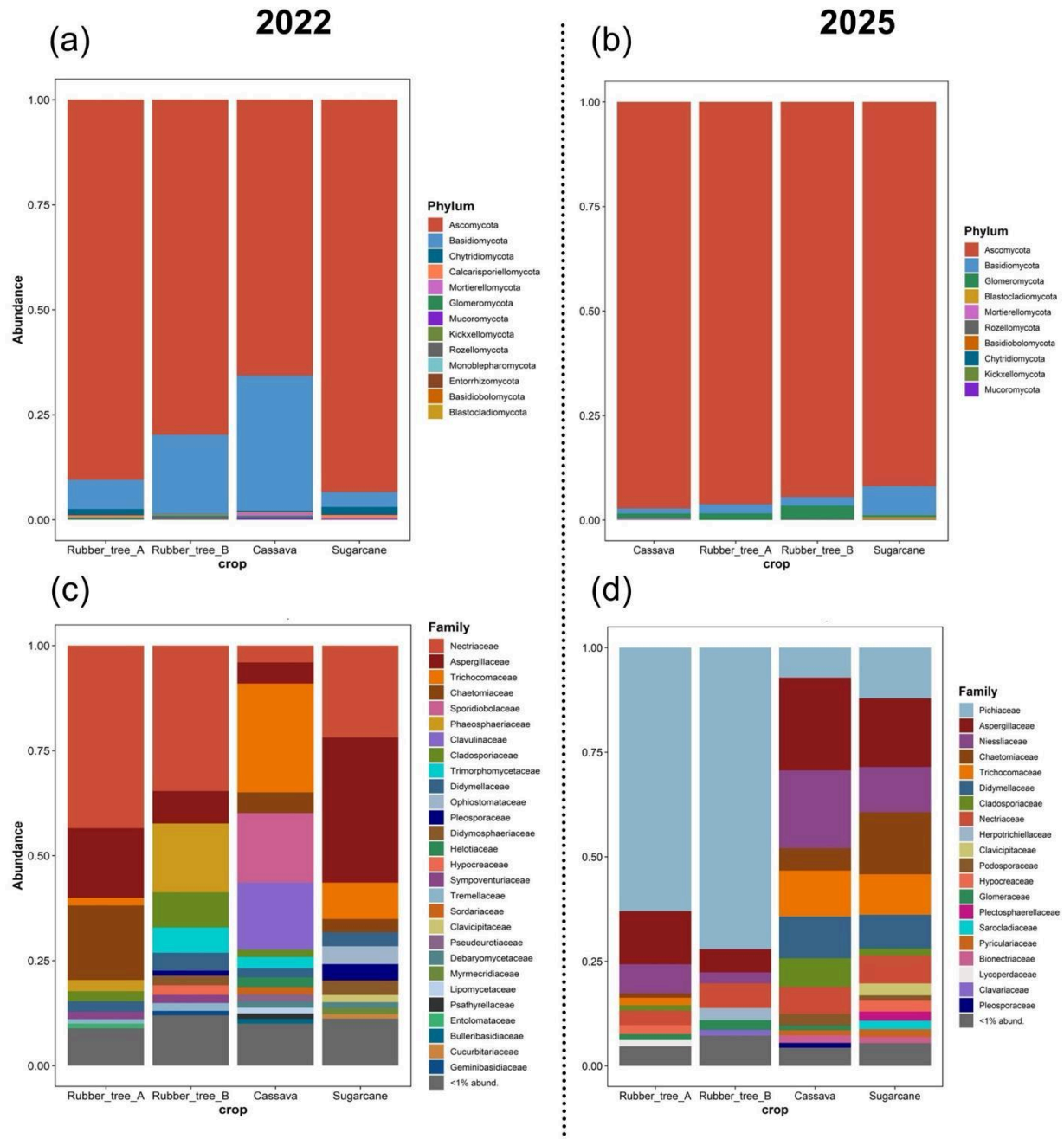


Figure S12. Taxonomic composition of soil fungal communities across four crop-associated systems in Sakon Nakhon in March 2022 and February 2025. Relative abundance bar plots are shown at multiple taxonomic levels. (a) Phylum- and (c) Family-level distribution in 2022; (b) Phylum- and (d) Family-level distribution in 2025. Only taxa exceeding 1% relative abundance are displayed, with remaining low-abundance taxa grouped as “<1% abundance”.

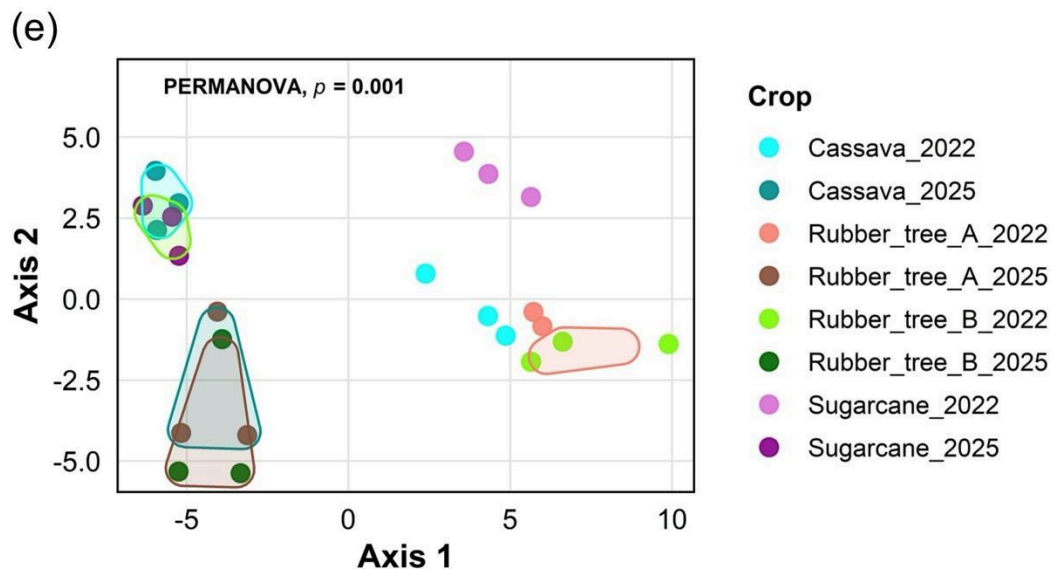
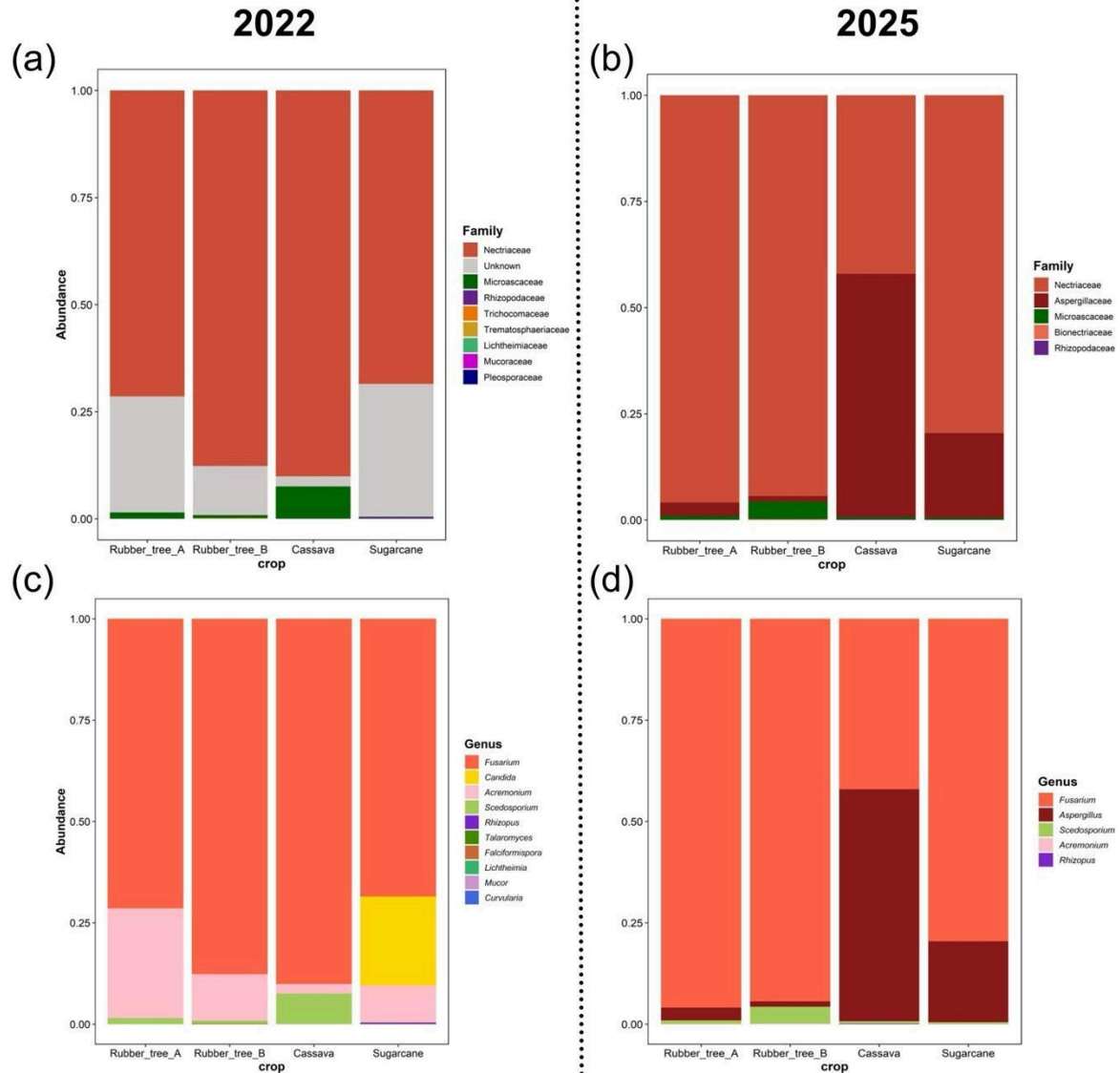


Figure S13. Relative abundance of WHO-FPPL taxa across four crop-associated soils in Sakon Nakhon in March 2022 and February 2025. Bar plots show taxonomic composition at the family and genus levels. (a) Family and (c) Genus--level distribution in 2022; (b) Family- and (d) Genus-level distribution in 2025; Only taxa exceeding 1% relative abundance are displayed, with remaining low-abundance taxa grouped as “<1% abundance. (e) Beta diversity ordination based on CLR-transformed Aitchison distances for taxa with abundance > 0, showing the combined analysis across both sampling years (PERMANOVA, $p = 0.001$).